

23E1011

IET, DAVV, Indore
B.E. 1 year (E&I)
Class Test – I (Oct.- 2023)
2AMRC1: Applied Mathematics-II

Time: 70 min.

Maximum Marks: 20

Note: Attempt only four questions.

Q:1 Define rank of matrix by normal form. Also convert the given matrix A to its normal form and (05)

hence find its rank where, $A = \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$.

Q:2 Examine whether the following system of equations is consistent. If so, find the solution: (05)

$$3x + 3y + 2z = 1; \quad x + 2y = 4; \quad 10y + 3z = -2, \quad 2x - 3y - z = 5$$

Q:3 Reduce the following equation into exact form and solve:- (05)

$$(1 + xy) y \, dx + (1 - xy) x \, dy = 0$$

Q:4 Solve the equation- $(D^2 - 3D + 2) y = x e^{3x} + \sin 2x$. (05)

Q:5 Solve- $p^2 + 2py \cot x = y^2$. (05)
